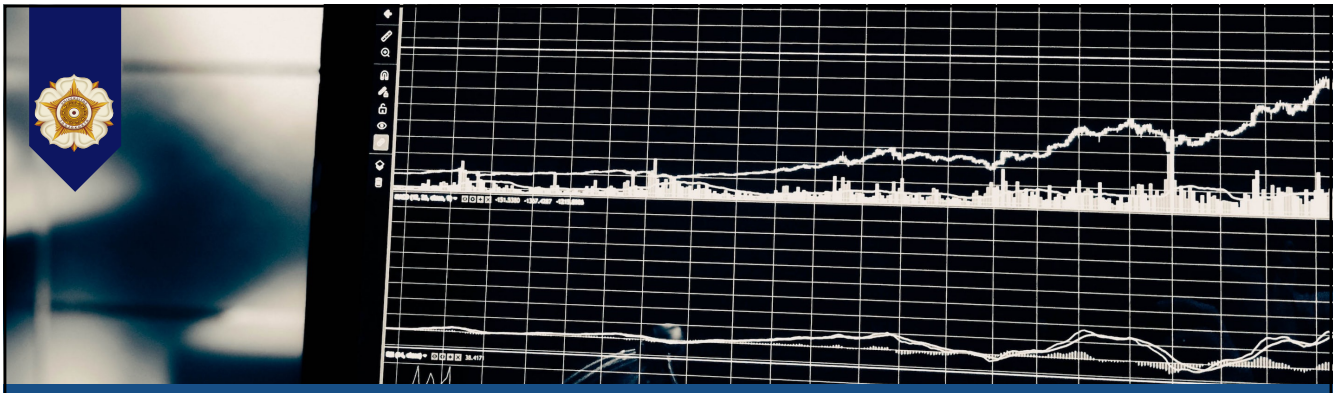


Introduction to Random Variable

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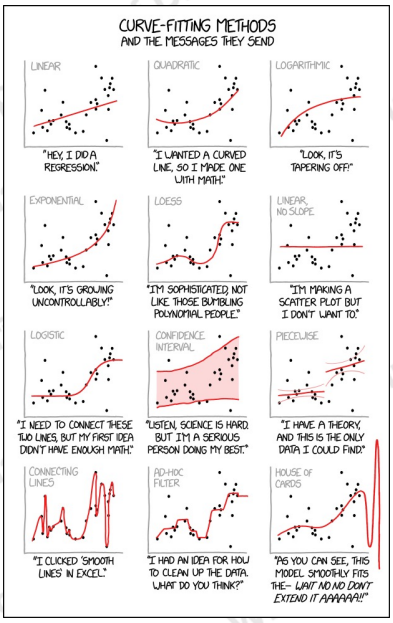
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1

Overview of today's lecture

- Random variable
- Discrete random variable
- Continuous random variable
- Joint probability and density function



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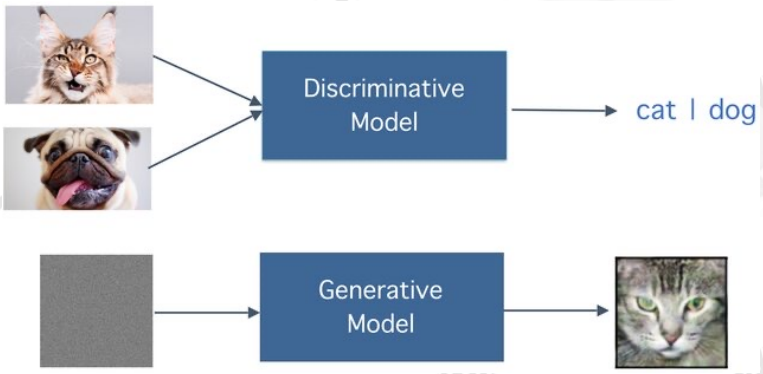
Part 1

RANDOM VARIABLE

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Probabilistic view of discriminative vs. generative model in AI



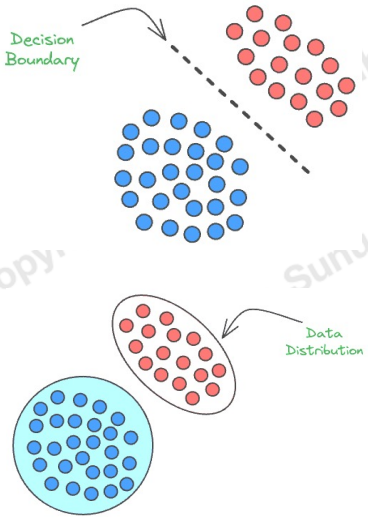
- Discriminative model learns to distinguish between two classes, i.e., pictures of cats or dogs.
- Generative model estimates the underlying distribution of a dataset (pictures of cats) and randomly generate realistic, yet synthetic, samples according to their estimated distribution.

https://www.researchgate.net/publication/341478640_An_Overview_of_Privacy_in_Machine_Learning

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Probabilistic view of discriminative vs. generative model in AI



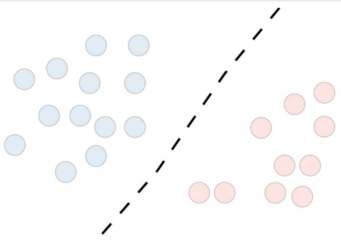
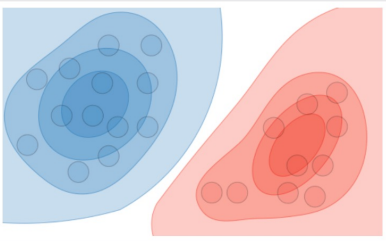
- **Discriminative models, also known as conditional models** focus on distinguishing between different classes or outcomes based on the input data. Their primary objective is to learn the decision boundary between classes.
- A discriminative model focuses on the conditional probability $P(y|x)$, where *y* represents the class label and *x* represents the input features: "*Given the features x, what is the probability of y being a specific class?*"
- **Generative models** are designed to learn how data can be generated. By understanding the joint probability distribution $P(x, y)$ of the features *x* and the class labels *y*, generative models can generate new data instances that are similar to the observed ones: "*Given the entire data distribution, how can we generate realistic instances of x for any given y, and vice versa?*"

https://generativeai.pub/machine-learning-employs-various-algorithms-to-interpret-data-learn-from-it-and-make-informed-1e898a25a7de

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Probabilistic view of discriminative vs. generative model in AI

	Discriminative model	Generative model
Goal	Directly estimate $P(y\|x)$	Estimate $P(x\|y)$ to then deduce $P(y\|x)$
What's learned	Decision boundary	Probability distributions of the data
Illustration		
Examples	Regressions, SVMs	GDA, Naive Bayes

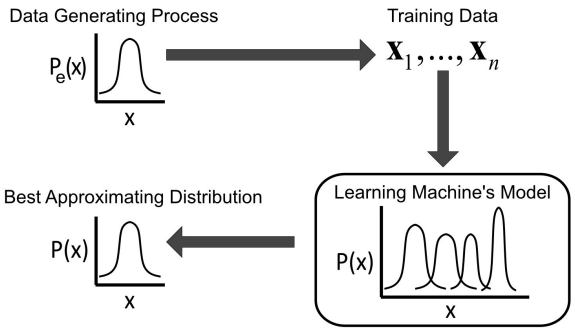
$$posterior = \frac{prior \times likelihood}{evidence} \Rightarrow P(Y|X) = \frac{P(Y) \cdot P(X|Y)}{P(X)}$$

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Why random variable and probability distributions?

- Some machine learning models work best under some distributions assumptions
- For example, these algorithms and function assume normal distribution: Linear Discriminant Analysis (LDA), Gaussian Naive Bayes, Logistic Regression, Linear Regression, Sigmoid function.
- When you are working with dataset, you are dealing **with sample instead of population**. Random variable and probability distributions help you to make predictions about the whole population.
- Understanding random variable and probability distributions help you to choose appropriate data transformation methods / feature extraction techniques.



- The Data Generating Process (DGP) generates observable training data from the unobservable environmental probability distribution *Pe*.
- The learning machine model observes the training data.
- Then, the model uses its beliefs about the structure of the environmental probability distribution in order to construct a best-approximating distribution *P* of the environmental distribution *Pe*.

Statistical Machine Learning: A unified framework by Richard M. Golden. Copyright 2021 by Richard M. Golden

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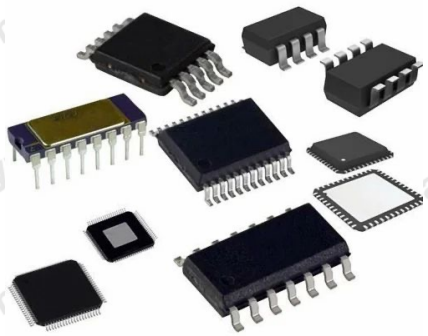
7

Random experiment

- Sometime, it is important to allocate a numerical description to the outcome.
- **Example:**
  The sample space giving a detailed description of each possible outcome when **three electronic components** are tested may be written:

  S ={NNN,NND,NDN,DNN,NDD,DND,DDN,DDD}

  where
  N: non-defective
  D: defective

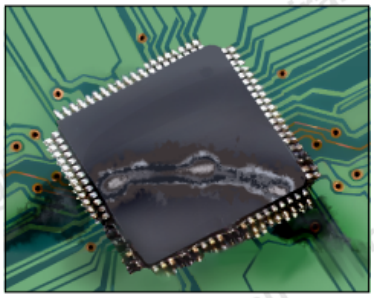


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Random experiment

- A supervisor is naturally concerned with **the number of defectives that occur (D)**.
- Thus, possible defective component (D) in the sample space will be assigned a numerical value of 0, 1, 2, or 3. S ={NNN,NND,NDN,DNN,NDD,DND,DDN,DDD}
- These values are random quantities determined by the outcome of the experiment (-i.e. we cannot judge whether we get 0, 1, 2, or 3 defective components in the outcome).
- They may be viewed as values assumed by the random variable X: the number of defective items when three electronic components are tested.



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What is random variable?

- A random variable X is a variable that assumes numerical values associated with events of an experiment.
- Use big letter *X* to denote random variables and its corresponding small letter *x* for one of its value
- Example: *X* = number of girls among 14 babies.

Table 4-1
Probabilities of Girls

x (girls)	*P*(*x*)
0	0.000
1	0.001
2	0.006
3	0.022
4	0.061
5	0.122
6	0.183
7	0.209
8	0.183
9	0.122
10	0.061
11	0.022
12	0.006
13	0.001
14	0.000

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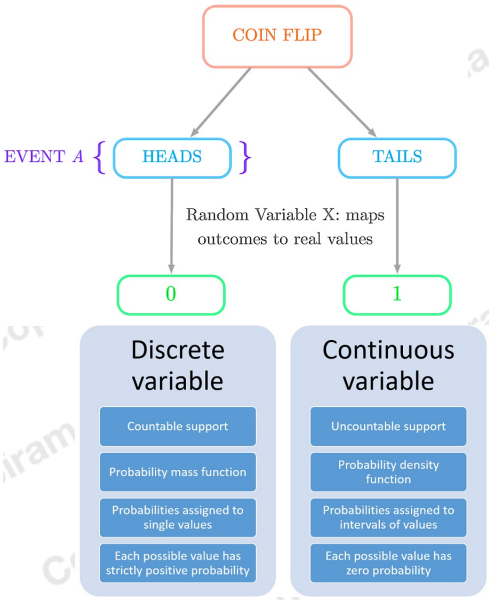
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Sample space

- **Discrete:**
  - A sample space containing a finite number of possibilities or an unending sequence with as many elements as there are numbers.
  - Variable called a "discrete random variable", can be counted
  - As shown in the figure, if you flip a coin, the two possible outcomes are 'heads' and 'tails'. An example of random variable would map 'heads' to 0 and 'tails to 1.
  - The event A corresponds to the following set of outcomes: {'heads'}. This means that the probability that the outcome is 'head' can be denoted as:

$$P(\mathrm{X}=0)=P('heads')=P(A)$$

- **Continuous sample space:**
  - Sample space containing an infinite number of possibilities equal to the number of points on a line segment.
  - Non-discrete
  - Variable is a "continuous random variable", cannot be counted but can be measured
  - Example: heights of children



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Part 2

DISCRETE RANDOM VARIABLE

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Discrete probability distributions

- Discrete random variable has a certain probability of equaling each of its possible values
  - Example: tossing coin 3 times
  - X = number of heads
  - S = {HHH,HHT, HTH, THH, TTH, THT, HTT, TTT}
  - For x=2, p=3/8
- Using formulae
  - f(x)= P(X=x)
  - f(3) = P(X=3) = 1/8



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Discrete probability distributions

- **The probability distribution** of a random variable is a function that takes **the sample space as input and returns probabilities**
- In other words, it maps possible outcomes to their probabilities (probability value of random variable X when its value equals to x)
- Set of ordered pairs (x, f[x]) is a probability function, or probability mass function, or probability distribution of the discrete random variable, X, if:

  $f(x) \geq 0$

  $\Sigma f(x) = 1$

  $P(X=x) = f(x)$

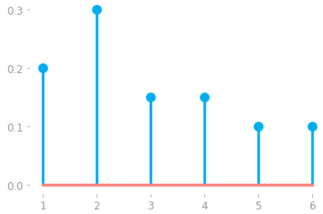


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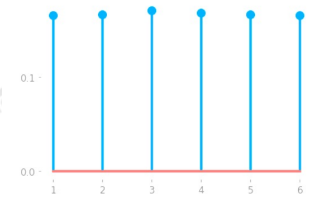
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Discrete probability distributions

- For instance, let's say that you're running a dice-rolling experiment.
- You call X the random variable corresponding to this experiment.
- Assuming that the die is fair, each outcome is ***equiprobable***.
- If you run the experiment **a large number of times**, you will get each outcome approximately the same number of times.



Probability mass function of the random variable X corresponding to a die rolling a six-sided die estimated from 20 rolls.



Probability mass function of the random variable X corresponding to a die rolling a six-sided die estimated from 100,000 rolls.

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Example of discrete probability distributions

Example 3.2: A stockroom clerk returns three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Solution: If S, J, and B stand for Smith's, Jones's, and Brown's helmets, respectively, then the possible arrangements in which the helmets may be returned and the number of correct matches are

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
JBS	0
BSJ	0

Correct match:

S	J	B

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Example of discrete probability distributions

Solution: If S, J, and B stand for Smith's, Jones's, and Brown's helmets, respectively, then the possible arrangements in which the helmets may be returned and the number of correct matches are

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
JBS	0
BSJ	0

m	0	1	3
$P(M = m)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

Note that the values of m exhaust all possible cases and hence the probabilities add to 1.

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Cumulative distributions for discrete probability distribution

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

For the random variable M, the number of correct matches in Example 3.2, we have

$$F(2) = P(M \leq 2) = f(0) + f(1) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

Sample Space	m
SJB	3
SBJ	1
JSB	1
JBS	0
BSJ	0
BJS	1

The cumulative distribution function of M is

$$F(m) = \begin{cases} 0, & \text{for } m < 0, \\ \frac{1}{3}, & \text{for } 0 \leq m < 1, \\ \frac{5}{6}, & \text{for } 1 \leq m < 3, \\ 1, & \text{for } m \geq 3. \end{cases}$$

One should pay particular notice to the fact that the cumulative distribution function is a monotone nondecreasing function defined not only for the values assumed by the given random variable but for all real numbers.

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Part 3

CONTINUOUS RANDOM VARIABLE

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What do you think?

- **Statement 1:**
  What is the probability of tomorrow morning temperature equals to 10° Celsius?
- **Statement 2:**
  What is the probability of tomorrow morning temperature less or more than 10° Celsius?



What is the answer of each statement?
Which statement is more appropriate to be used?

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Continuous probability distributions

- Called: **Probability Density Function (PDF)**
- **Continuous** random variable has P = 0 of being ***exactly*** **any value**.
- Example: probability that a person is *exactly* 175 cm tall
  - Prob (Height = 175.00000000000)=0
- However, we *can* find probability that the height lies within some range
  - example: $p(\text{Height} \geq 175)$ or $p(160 < \text{Height} < 175)$

Remember: probability of continuous random variable must be described over a range

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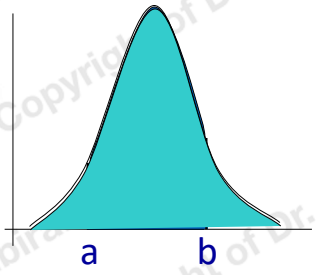
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Continuous distributions

$$P(a < X < b) = \int_a^b f(x)\,dx$$

P(a < X < b) must be positive

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$



Sum of all f(x) must equal unity

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Continuous distributions

$$P(a < X < b) = \int_a^b f(x)\,dx$$

$$P(a < X < b) = P(a \le X \le b)$$

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Review for practical purpose

INTEGRALS

Assume (u, v) are functions of x and (a, n) are constants. An integration constant C should be added to each indefinite integral.

$$\int a\,dx = ax \qquad \int u\,dv = uv - \int v\,du$$

$$\int ax^n\,dx = a\frac{x^{n+1}}{n+1}, n \neq -1 \qquad \int \frac{a}{x}dx = a\ln|x|$$

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax \qquad \int \sinh ax\,dx = \frac{1}{a}\cosh ax$$

$$\int \cos ax\,dx = \frac{1}{a}\sin ax \qquad \int \cosh ax\,dx = \frac{1}{a}\sinh ax$$

$$\int \tan ax\,dx = -\frac{1}{a}\ln|\cos ax| \qquad \int \tanh ax\,dx = \frac{1}{a}\ln(\cosh ax)$$

$$\int e^{ax}dx = \frac{1}{a}e^{ax} \qquad \int \ln ax\,dx = x\ln ax - x$$

$$\int x\sin ax\,dx = \frac{1}{a^2}\sin ax - \frac{x}{a}\cos ax \qquad \int x\cos ax\,dx = \frac{1}{a^2}\cos ax + \frac{x}{a}\sin ax$$

$$\int \sin^2 ax\,dx = \frac{x}{2} - \frac{\sin 2ax}{4a} \qquad \int \cos^2 ax\,dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$$

$$\int \sinh^2 ax\,dx = \frac{\sinh 2ax}{4a} - \frac{x}{2} \qquad \int \cosh^2 ax\,dx = \frac{\sinh 2ax}{4a} + \frac{x}{2}$$

$$\int x^n e^{ax}dx = \frac{1}{a}x^n e^{ax} - \frac{n}{a}\int x^{n-1}e^{ax}dx \qquad \int x^n \ln ax\,dx = \frac{x^{n+1}}{n+1}\ln ax - \frac{x^{n+1}}{(n+1)^2}$$

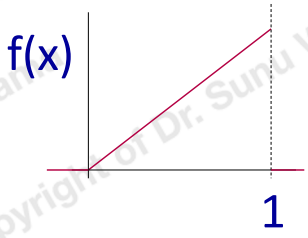
$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} \qquad \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|$$

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Example

- Given that a random variable, X, has a density function:
  - f(x) = 2x, for 0 < x < 1
  - f(x) = 0, for all other x
- Verify area under curve = 1.0



$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{0} 0\,dx + \int_{0}^{1} 2x\,dx + \int_{1}^{\infty} 0\,dx$$

$$= \frac{2x^2}{2}\Big|_0^1 = x^2\Big|_0^1 = 1^2 - (0)^2$$

= 1.0

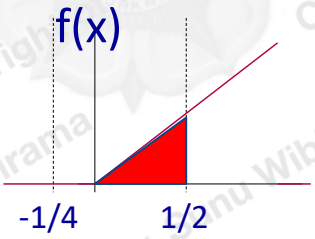
$$\int ax^n\,dx = a\frac{x^{n+1}}{n+1}, n \neq -1$$

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Example

- Given that a random variable, X, has a density function:
  - f(x) = 2x, for 0 < x < 1
  - f(x) = 0, for all other x
- What is the probability that: -¼ < x < 1/2



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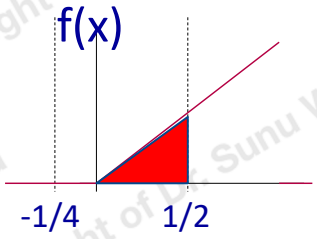
Answer

- Given that a random variable, X, has a density function:
  - f(x) = 2x, for 0 < x < 1
  - f(x) = 0, for all other x
- What is the probability that: -¼ < x < 1/2

$$P(-¼ < x < ½) = \int_{-1/4}^{0} 0\,dx + \int_{0}^{1/2} 2x\,dx$$

$$= x^2 \Big|_{0}^{1/2} = (1/2)^2 - (0)^2$$

= ¼ - 0 = ¼

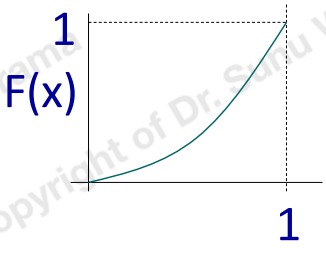


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Cumulative distributions for continues random variable

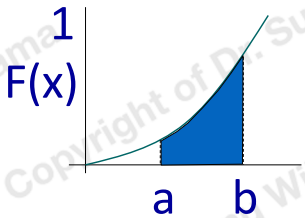
- The cumulative distribution, F(x) of a continuous random variable, x, with density function, f(x), is:

$$F(a) = P(X \le a) = \int_{-\infty}^{a} f(x)\,dx$$



P(a<x<b) = F(b) - F(a)

$$= \int_{-\infty}^{b} f(x)\,dx - \int_{-\infty}^{a} f(x)\,dx = \int_{a}^{b} f(x)\,dx$$



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Differentiation of F(x)

$$f(x) = \frac{dF(x)}{dx}$$

Lets you determine f(x) from F(x)

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Example

- Suppose the error in the reaction temperature for a experiment is a continuous random variable, x, having the cumulative probability density function of:

  $F(x) = x^3/9$, for $-1 < x < 2$
  $F(x) = 0$, for all other $x<-1$ and $x>2$

  - Find expression for density function f(x) for $(-1< x < 2)$

$$f(x) = d[F(x)]/dx$$
$$= d[x^3/9]/dx$$
$$= x^2/3$$

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Example

Find F(x) if we have: $f(x) = x^2/3$, for $-1 < x < 2$
$f(x) = 0$, for all other x

$$\int ax^n\,dx = a\frac{x^{n+1}}{n+1}, n \neq -1$$

$$F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{-1}^{x} \frac{t^2}{3}\,dt = \frac{t^3}{9}\Big|_{-1}^{x} = \frac{x^3+1}{9}$$

- Therefore:

$$F(x) = \begin{cases} 0 & x \le -1 \\ \dfrac{x^3+1}{9} & -1 \le x < 2 \\ & x \ge 2 \end{cases}$$

What's here?

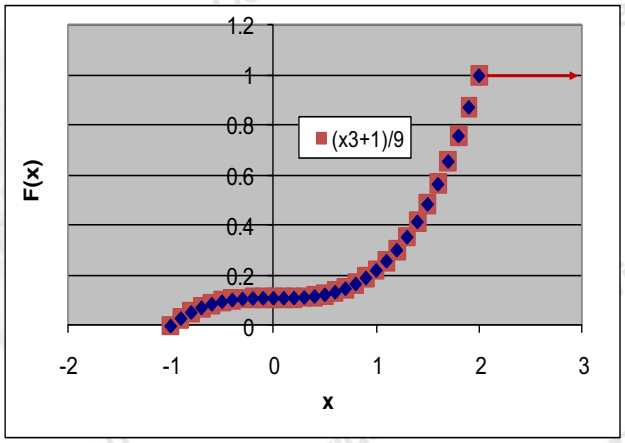
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Example

$$F(x) = \begin{cases} 0 & x \le -1 \\ \dfrac{x^3+1}{9} & -1 \le x < 2 \\ 1 & x \ge 2 \end{cases}$$

Should monotonically approach 1

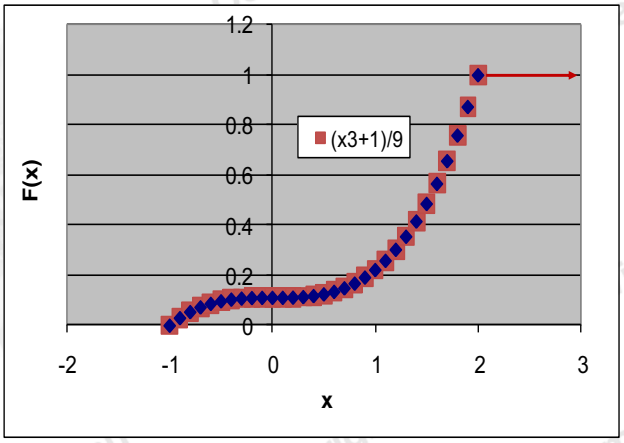


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Example

$$F(x) = \begin{cases} 0 & x \le -1 \\ \dfrac{x^3+1}{9} & -1 \le x < 2 \\ 1 & x \ge 2 \end{cases}$$



Compute:

$P(0 < x \le 1) = F(1) - F(0)$

$= 2/9 - 1/9 = 1/9$

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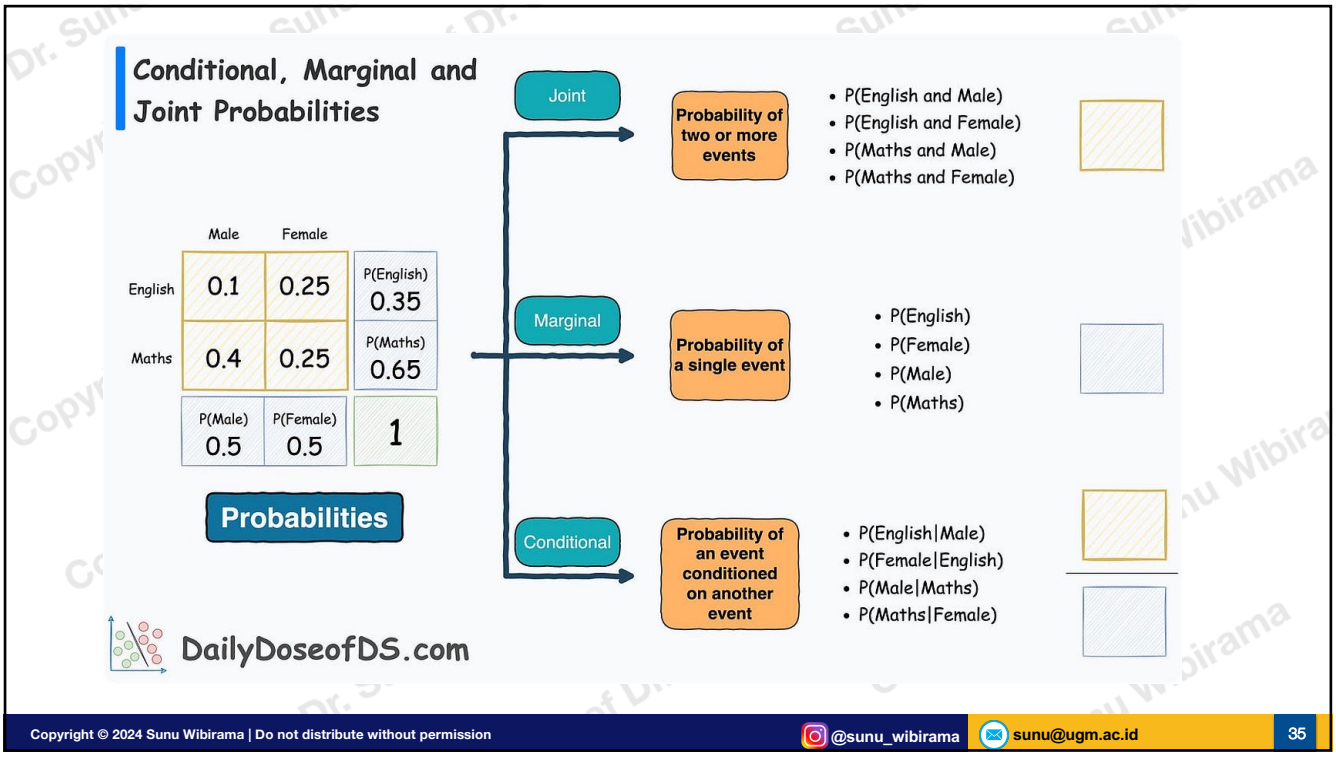
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Part 4

JOINT PROBABILITY AND DENSITY FUNCTIONS

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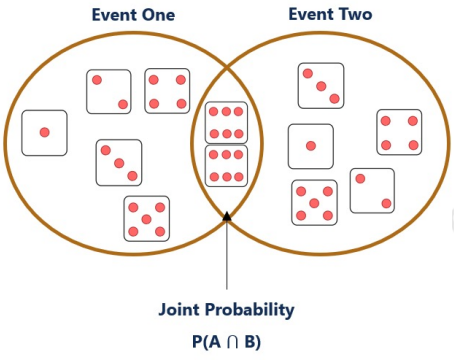


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Joint probability: a review

- **Joint probability:** is a statistical measure used to calculate the likelihood of two events taking place at the same time.
- Both events must be independent of each other.
- Joint probability is also called the intersection of two or more events.
- It is different from conditional probability, which refers to the probability that one event will happen when another event takes place.



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Joint probability distributions

[discrete random variables]

- Joint probability distribution: may have several random variables of interest.
- Function f(x, y) is a joint probability distribution of the discrete random variables X and Y if:

$$f(x,y) \geq 0, \quad \text{for all } (x,y)$$

$$P(X= x, Y= y) = f(x,y)$$

$$\sum_x \sum_y f(x,y) = 1$$

Discrete variable

- Countable support
- Probability mass function
- Probabilities assigned to single values
- Each possible value has strictly positive probability

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Example

Two refills for a ballpoint pen are selected at random from among 3 blues, 2 reds, and 3 greens. If X is the number of blue refills and Y is the number of red refills selected, find the joint probability function f(x,y), {(x,y)|x+y <=2}

Solution:

Possible pairs of numbers of blue & red are: (0,0), (0,1), (1,0), (1,1), (0,2), (2,0)

(0,0) means : no blue, no red (but there is possibility that two green pens are selected)

(0,1) means: no blue, 1 red (but there is possibility that one green pen is selected)

(1,1) means : 1 red, 1 blue

Now, $f(0,1)$, for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$. The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1}\binom{3}{1} = 6$. Hence, $f(0,1) = 6/28 = 3/14$. Similar calculations yield the probabilities for the other cases, which are presented in Table 3.1. Note that the probabilities sum to 1.



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Example

Let's represent the computation in formula:

$$N_{total\ combinations} = \binom{8}{2} = \frac{8!}{2!(8-2)!} = \frac{8(7)(6!)}{2(6!)} = 28$$

$$N_{(r,b,g)} = \left(\frac{3!}{r!(3-r)!}\right)\left(\frac{2!}{b!(2-b)!}\right)\left(\frac{3!}{g!(3-g)!}\right)$$

$$N_{(r,b)} = \left(\frac{3!}{r!(3-r)!}\right)\left(\frac{2!}{b!(2-b)!}\right)\left(\frac{3!}{(2-r-b)!(3-[2-r-b])!}\right)$$

$$= \left(\frac{3!}{r!(3-r)!}\right)\left(\frac{2!}{b!(2-b)!}\right)\left(\frac{3!}{(2-r-b)!(1+r+b)!}\right)$$

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Example

f(r,b)	r = 0	r = 1	r = 2	h(b)
b = 0	3/28	9/28	3/28	15/28
b = 1	6/28	6/28		12/28
b = 2	1/28			1/28
g(r)	10/28	15/28	3/28	1

$$f(r,b) = \left(\frac{3!}{r!(3-r)!}\right)\left(\frac{2!}{b!(2-b)!}\right)\left(\frac{3!}{(2-r-b)!(1+r+b)!}\right)$$

Find P(r,b) of A, where A={ (r,b)|(r+b) ≤ 1}

= f(0,0) + f(0,1) + f(1,0)

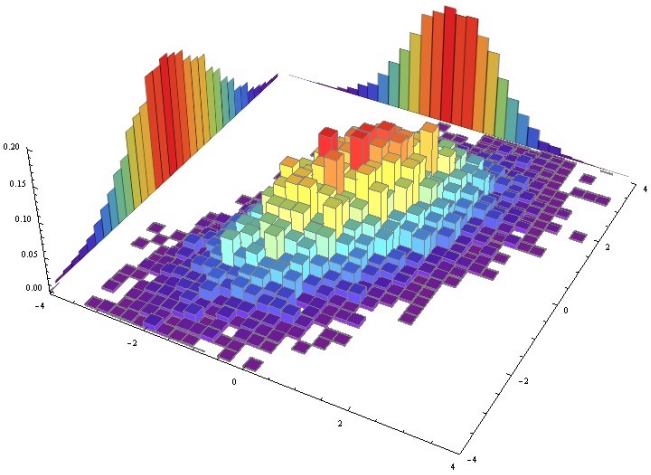
= 3/28 + 6/28 + 9/28

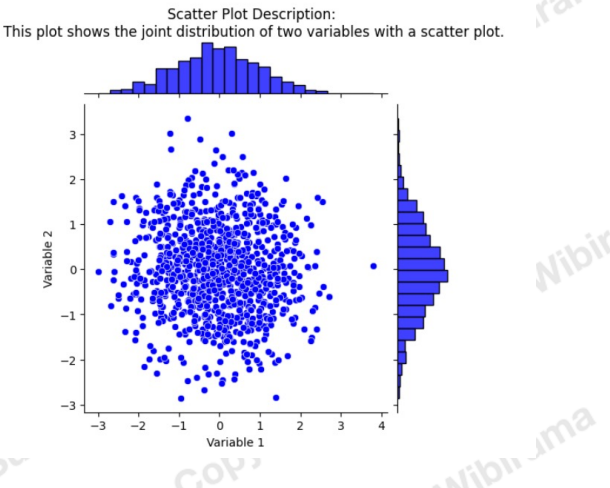
= 18/28

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Visualization of joint probability distribution





https://stackoverflow.com/questions/51306907/visualization-of-bivariate-probability-distribution

https://medium.com/@ebimsv/ml-series-day-30-joint-distribution-marginal-distribution-and-conditional-distribution-2794b600680b

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Joint density function

[continuous random variables]

- The function, f(x,y), is a joint density function of the continuous random variables X and Y if:

$$f(x,y) \geq 0, \quad \text{for all } (x,y)$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

$$P[(X,Y) \in A] = \iint_{A} f(x,y)\,dx\,dy$$

Continuous variable

- Uncountable support
- Probability density function
- Probabilities assigned to intervals of values
- Each possible value has zero probability

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Example

- A candy company produces boxes of chocolates with a mixture of creams, toffees, and nuts coated in both light and dark chocolates.
- For a randomly selected box, let X= weight of light creams and Y = weight of dark creams. The joint density function is:

f(x,y) = 0.4 (2x + 3y), $0 \le x \le 1$, $0 \le y \le 1$

= 0 for all other ranges

- Verify that: $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

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Example

- The joint density function is:

f(x,y) = 0.4 (2x + 3y), $0 \le x \le 1$, $0 \le y \le 1$

= 0 for all other ranges

- Verify that: $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$
- Answer :

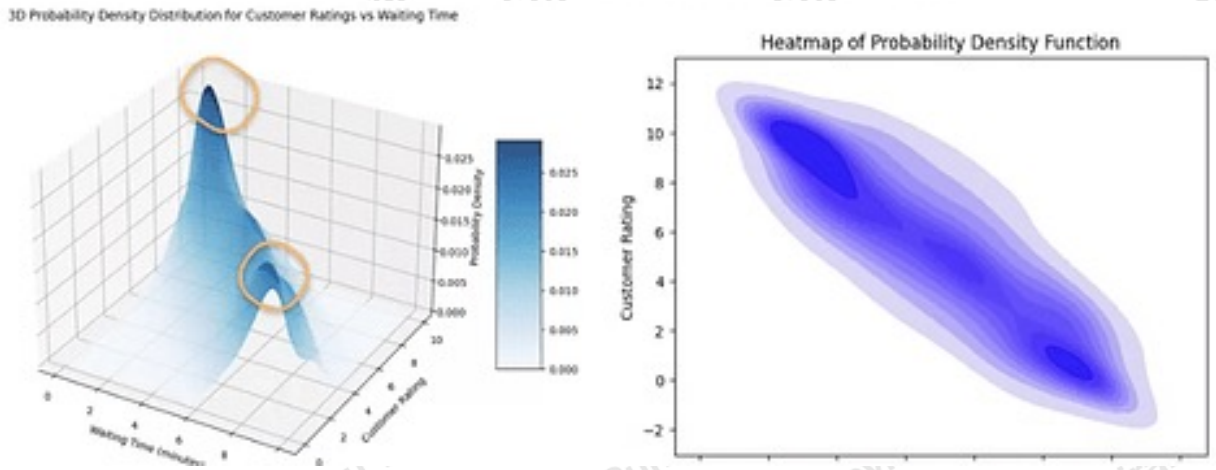
$$\int_0^1\int_0^1 0.4\,(2x+3y)\,dx\,dy = 0.4\int_0^1 (x^2+3xy)\Big|_0^1\,dy$$

$$0.4\int_0^1 (1+3y-0)\,dy = 0.4[y+1.5y^2]\Big|_0^1 = 0.4[1+1.5-0] = 1$$

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Visualization of joint density function



https://community.deeplearning.ai/t/missing-explanation-joint-distribution-continuous/390333

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Take home messages

- In this lecture, we learn about random variable. A random variable X is a variable that assumes numerical values associated with events of an experiment.
- The probability distribution of a random variable is a function that takes the sample space as input and returns probabilities
- There are two types of random variable: discrete and continuous. Each type needs different formula to compute its probability distribution.
- If we want to compute the distribution of more than one random variable, we need to compute joint probability distribution (discrete RV) or joint density function (continuous RV).

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